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Properties of Laser Light

Student Notes

Contents

1 Experimental Aim

To investigate the characteristics of a low-powered 'Melles Griot' (Class 2) Helium Neon (He-Ne) Laser at visible red light (633nm).

2 Introduction

This experiment is about the properties peculiar to the light emitted by lasers, in particular the **Helium-Neon (He-Ne) laser**, and the relationship between these properties and the workings of a laser. A basic understanding of how a He-Ne laser works can be gained from *Ref.1 a) pg.596-597*. To start off, students should begin reading from *pg.586* (excluding the section on "The First (Pulsed Ruby) Laser").

The laser used for this experiment is of the "unpolarized" or "randomly polarized" variety since it has no Brewster windows. The mirrors, which form a quasi-hemispherical resonant cavity, are sealed directly onto the ends of the plasma tube. More information about the various types of laser cavities and their applications can be found in *Ref.1 a) and 2*.

3 References

1. **a) Chapter 13 Modern Optics: Lasers and Other Topics pg.586-597.**
   b) Chapter 12 Basics of Coherence Theory pg.560-573.
   Optics (4th Edition) by E. Hecht, Addison Wesley.

2. **Chapter 5 Lasers I**
   Optoelectronics - an introduction (3rd Edition) by J. Wilson and J. Hawkes, Prentice Hall, pg.169-195.

3. **Chapter 3 Beam Optics - 3.1 The Gaussian Beam**
   Fundamentals of Photonics (2nd Edition) by B.E.A. Saleh and M.C. Teich, Wiley, pg.74-86. This book can also be accessed online through the UNSW Library website.

4. **Spatial Filtering**
   "Extract from "Optics Guide 4", Melles Griot (1988)

5. **Visual Effects with Laser Light**
   Laser Light: Fundamentals and Optical Communication by C.C. Eaglesfield, Macmillan, pg.179-182.

6. **Polarization and intensity fluctuations in He-Ne lasers**
   E. Fortin and D. Singh, Am. J. Phys. 49 891 (1981)

4 Experiment

Precautions

- Do not look directly into the laser (or a specular refection of it).
- The eye's natural blink reflex is enough to prevent permanent damage should you accidentally view the beam directly.
- Make sure that when sitting or standing your eye level is above the laser beam level. This will minimize the risk of accidentally viewing the laser beam directly.

4.1 Properties of LASER Light - Basics

4.1.1 Experiment 1 - Wavelength of Red Light

Reference 1 a) pg.596-597

Aim: *To estimate the wavelength of the neon red line.*

Method:

1. Turn on the laser if you have not already done so.
2. Find the diffraction grating which can be found in a small red box marked as "grating".

   Hold the grating only by the edges so as not to damage it!
3. Mount the grating in the slide holder (apparatus with a black window and 2 metal clips) on the optical rail so that the grating lines are vertical.
4. Slide the grating along the optical rail until it is approximately 60cm away from the black screen found at the end of the bench and align the laser to the grating. Make sure that both the zeroth and first order maxima appear on the screen so that their separation can be easily measured.
5. Measure (including errors) (1) the grating to screen distance and (2) the separation between the undeviated beam and the first order diffraction maximum. Assume a right-angled triangle is formed between the grating, screen and diffraction maxima.

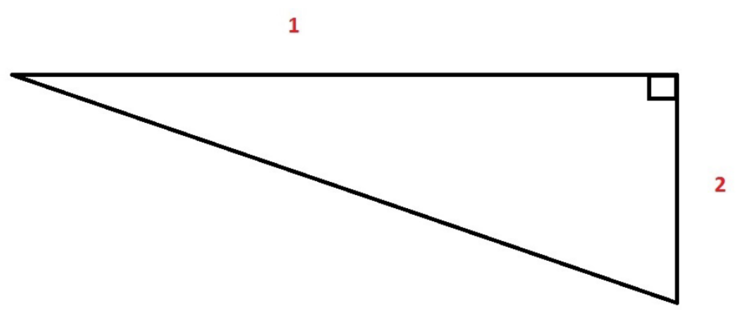

Figure 1: *Diagram of the diffraction grating setup to estimate the wavelength of the red He-Ne laser.*

Questions

1. Calculate the angle of diffraction and hence your measured value of the wavelength of red light (with error) using the equation $d \sin\theta = n\lambda$. Compare with the accepted value of 633nm. Note that the grating spacing $d = 1.69\mu$m.

4.1.2 Experiment 2 - Spectroscopy: Colours Emitted

Aim: *To measure and investigate the emitted spectra from the He-Ne Laser, Helium, Neon discharge tubes and the effect the laser cavity mirrors have on the laser spectrum.*

Method:

The spectrometer probe (long blue cable) connected to the computer via the "Spectra-Suite" software will be used to measure the spectra of six different light sources. In your lab book, you will present the spectra in 3 pairs to facilitate comparisons. In each spectrum the 633nm wavelength should be clearly indicated with the cursor.

Laser output vs. Laser cavity (He-Ne plasma)

1. From the desktop, open the "SpectraSuite" software.
2. Click on the "Play" button to start detection.
3. Point the spectrometer probe at a diffuse reflection of the He-Ne laser to measure its spectrum.

   DO NOT try and measure the spectrum by pointing the probe directly in line with the laser output as this can damage the spectrometer.
4. Observe the spectra on the screen and capture it by clicking the "Pause" button. Take care not to saturate the spectrum by making sure any observed peaks are not "clipped" (maxima having flat tops). If this is occurring, simply move the probe further away from the diffuse reflection to reduce the detected intensity. This applies to all spectra measured.
5. Click anywhere on the graph and a vertical cursor will appear. Set this cursor to 633nm for ALL the spectra you take.
6. Save your He-Ne laser spectrum by clicking the "Save" button. A window will pop up where you will name your spectrum. For "File Type", select "Tab Delimited" and click "Save".
7. Import the data from your saved text file in to a program of your choice e.g. Excel to plot the spectrum. Do this for all subsequent spectra measured.
8. Now take the Helium and Neon plasma spectrum by pointing the spectrometer probe into the tiny hole found on the side of the laser (remove the sticker covering the hole).

Helium Plasma vs Neon Plasma

1. Locate the "LLE-3 Multiple Discharge Lamp" which can be found to the right of the computer.
2. To take the Helium and Neon plasma spectra, point the spectrometer probe at the corresponding discharge tube. To switch between plasma, insert the selector in the corresponding hole.

Incandescent Light Bulb vs. Incandescent Light Bulb through Laser Tube

1. Locate the black desk lamp which can be found to the left of the computer.
2. To take the spectrum of the incandescent light bulb, point the spectrometer probe at the bulb.
3. To take the incandescent light bulb spectrum through the laser tube, use a laser tube that is not inside an operating laser (ask the demonstrator for help if you can't find it). Make sure to measure the spectra through the dielectric mirrors on the ends of the tube.

Questions

Note: You can skip these questions and continue to Experiment 3 - Beam Divergence during your lab session.

Based on the spectra you have just recorded and using ref 1a) and 2, answer the following questions to better understand how a He-Ne works:

1. What is the wavelength of the laser output?
2. Is the laser output line in the laser cavity spectrum? Is it a major peak? Identify this peak in the appropriate He or Ne spectra.
3. Where do the rest of the laser cavity peaks come from? Identify the 5 most intense peaks in the appropriate He or Ne spectra. You can look up the lines and corresponding level transitions on the NIST Atomic Spectra Database.

   `physics.nist.gov/PhysRefData/ASD/lines_form.html`

4. Briefly describe how the He-Ne laser works and list some of the sources of cavity losses.
5. What is the effect of the laser tube on the incandescent lamp spectra? How does this relate to the reflectivity spectrum of the laser mirrors?
6. Why are the mirrors highly reflective at 633nm? How does the red light get out of the laser cavity?
7. What are the 2 necessary conditions for lasing action? From this, explain why out of all the He and Ne spectral lines, only the 633nm line is amplified.

4.1.3 Experiment 3 - Beam Divergence

Reference 3 pg.74-86

Aim: *To measure the approximate divergence of the laser beam.*

Method:

1. Find the Spiricon Beam Profiler and connect it via USB to the computer.
2. Mount it onto the optical rail using the translational stage, perpendicular to the beam and align the laser beam onto the beam profiler.
3. From your desktop, open the "LabVIEW" program "PLL 3 - Beam Divergence".
4. To start the program, click the "Run" button (white arrow in the top left). A window prompting for your name will appear. Enter your name, select where you wish to save your data then click "OK". The "BeamGauge" software should then initialise in the background.
5. Once the software has initialised (the "Ready" button should be visible) click the "CLICK TO START ACQUIRING DATA" button. A profile of your laser beam should now appear. Using the stage controls, adjust the beam so that it is centered in the image box. If the detector is saturated, reduce the exposure by dragging the exposure marker to the left. The maximum pixel intensity should be less than 100.
6. Starting at a distance of 10cm away from the laser aperture, record the FWHM of the beam profile. To do this, select the line feature beside the image box and draw a line through the centre of your beam profile by clicking and dragging. The experimental data should have a gaussian form. In LabVIEW, record the distance at which the FWHM was taken. Finish by clicking the "ADD DATA POINT TO PLOT" button.
7. Repeat the measurement by recording the FWHM in intervals of 2cm from 10-30cm, then 5cm intervals between 30-130cm. Once you have recorded all your measurements, a plot of the FWHM vs Distance from the laser aperture will be created.
8. To save your data and graph, right click on your beam divergence graph and hover over "Export" to find the relevant selections. Insert your graph and data into your lab book.

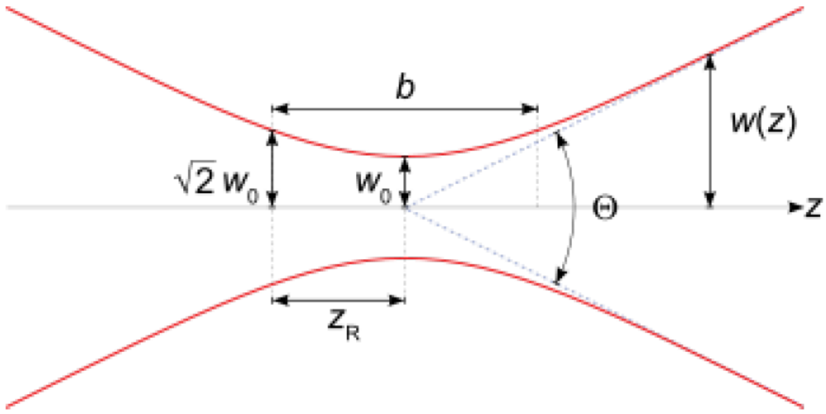

Figure 2: *Beam Divergence*

Questions

1. From your graph, calculate (1) the divergence of the laser (with error) in milliradians and (2) the beam diameter at the laser exit aperture. On your graph, label the beam's Rayleigh Range.

   Note: Divergence is approximately equal to:
   change in beam diameter / change in distance from the laser aperture.

2. Given that the beam diameter at the laser aperture is 0.88mm as quoted by the manufacturer, calculate the theoretical minimum divergence of the laser beam. Compare this with the minimum divergence of the laser beam using your results.

4.1.4 Experiment 4 - Spatial Filter

Reference 4

Aim: *To find the pinhole diameter that is best suited to the spatial filtering of a laser focused by a given microscope lens.*

In this experiment, you will learn how to use a spatial filter (pinhole) to "clean" up the laser's intensity distribution. Smaller spatial filters are able to clean up the beam more effectively, but result in a loss of laser intensity. A balance between beam cleanness and laser power needs to be met.

Method:

1. Keeping the Spiricon beam profiler on the optical rail from the previous experiment, move it to a distance of approximately 30cm from the laser aperture.
2. From the desktop, open the LabVIEW program "PLL 4 - Spatial Filter".
3. To start the program, click the "Run" button (white arrow in the top left). A window prompting for your name will appear. Enter your name, select where you wish to save your data then click "OK". The "BeamGauge" software should then initialise in the background.
4. Once the software has initialised (the "Ready" button should be visible) click the "CLICK TO START ACQUIRING DATA" button. A profile of your laser beam should now appear. Using the stage controls, adjust the beam so that it is centred in the image box. If the detector is saturated, reduce the exposure by dragging the exposure marker to the left. The maximum pixel intensity should be less than 100.

   Maintain this exposure level throughout the experiment.
5. Find the spatial filter assembly "Newport Model M-900" and mount it on the optical rail in between the laser and the beam profiler. Remove any spatial filters currently placed in the assembly.
6. Using the controls on the assembly, adjust the microscope objective so that the laser passes through its centre. Check this by using a piece of paper. Once finished, you should be able to view the beam in LabVIEW. The beam spot on the profiler should be in the relatively same position as when no objective is used. Some re-alignment of the laser may be needed. Ask a demonstrator if you need help.
7. In LabVIEW, select the "no pinhole" option in the "Current Line Profile" box.
8. Draw a line through the centre of your beam image. An intensity distribution will be seen.
9. To save your data, click the yellow "SAVE DATA" button. An image of your plot as well as 2 text files giving the intensity data and power through the spatial filter as a percentage of no pinhole power (100%) should be produced.

10. Now select the "100 microns" option in LabVIEW and insert the 100μm spatial filter in the assembly. Using the controls, align the spatial filter so that the beam passes through it. See the section "How to Align a Spatial Filter" at the end of this experiment. Aim to maximise the power through spatial filter.
11. Draw a new line through the centre of your beam image if required then save your data. A window will pop up asking if you want to replace the existing saved files. Click "Yes". Your plots and data will be combined with any existing data
12. Repeat the measurements with the 50μm and 25μm spatial filters. Make sure to select which line profile you are using in LabVIEW before inserting the corresponding spatial filter in the assembly. Insert your plots and power data into your lab book.

How to Align a Spatial Filter

As the size of the spatial filter pinhole is small, getting the laser beam to pass through it effectively can be tricky. A systematic method to do this is as follows:

1. Place the spatial filter in the assembly.
2. Use the assembly z-axis (parallel to the optical rail) control to wind the microscope objective as far away as possible from the spatial filter.
3. Then using the assembly spatial filter position controls, align the pinhole through the beam. Check with a piece of paper that the beam through the pinhole is maximised.
4. Now move the objective slightly closer to the spatial filter then re-adjust the spatial filter position so that the beam is again maximised. Repeat this step until you find that the maximum intensity through the spatial filter is reached. This should be when the spatial filter coincides with the focal length of the objective. Ask a demonstrator for help if you are having difficulties.

Questions

1. From your 4 plots and the power achieved through each spatial filter, determine which is most useful.
2. Calculate the optimum pinhole diameter which is given by $d = \pi w_0 = \frac{\lambda f}{w}$ where f = 30mm is the focal length of the objective. A pinhole diameter can allow 99.3% of the laser power through. Compare this value with the pinhole you chose to be most useful.

4.1.5 Experiment 5 - Granularity of Scattered Light

Reference 5

Aim: *To observe and explain some phenomena associated with the scatter of spatially coherent light.*

Using the $100\mu m$ spatial filter, expand the laser beam and project it onto the wall at the end of the bench to give a beam spot of about 10cm in diameter. Complete the following exercises and write down your observations in your lab book.

Method:

1. Keeping your head very still observe the illuminated area from distances of about 1 and 3 metres and notice how the size of the speckle is affected by the wall-eye separation.
2. With your eye about 1 metre from the wall observe the speckle through a range apertures (piece of metal labelled "Apertures") from 0.8mm to 1.6mm and note how the aperture size affects the size of the speckle.
3. Now, observe the speckle pattern from about 2 metres away with one eye closed. Slowly move your head sideways and note whether the speckle motion is in the same or opposite sense to your movements. Repeat with the other eye.
4. Now, observe the speckle pattern from about 2 metres away with one eye closed. Slowly move your head sideways and note whether the speckle motion is in the same or opposite sense to your movements. Repeat with the other eye.
5. Look at the speckle pattern through:

   (a) a converging lens (blue, red or yellow) (this simulates myopia, or short-sightedness) and note the sense of speckle motion as you move your head sideways.

   (b) a diverging lens (brass) (this simulates hyperopia, or long-sightedness) and note the sense of the speckle motion as you move your head sideways.

6. Draw a table in your lab book to summarize all your observations about the speckle size and direction of movement. Explain your observations.

4.2 Properties of LASER Light - Advanced

4.2.1 Experiment 6 - Laser Stability

Reference 6

Aim: *To monitor the power output of the laser as a function of time after switching the laser on. To monitor the polarisation of the laser beam as a function of time.*

**** DO NOT switch on the laser yet! ****

Method:

Unpolarised warm-up curve

1. Set up the Spiricon beam profiler so that it is approximately 30cm away from the laser on the optical rail. Align the laser with the beam profiler.
2. From the desktop, open the LabVIEW program "PLL 6 - Warm Up".
3. To start the program, click the "Run" button (white arrow in the top left). A window prompting for your name will appear. Enter your name, select where you wish to save your data then click "OK". The "BeamGauge" software should then initialise in the background.
4. Once the software has initialised (the "Ready" button should be visible) click the "BEAM PROFILER OFF (CLICK TO TURN ON)" button. If the detector is saturated, reduce the exposure by dragging the exposure marker to the left. The maximum pixel intensity should be less than 100.
5. Turn on your laser then immediately click the "CLICK TO START RECORDING DATA" button with your data acquisition time set at 15 minutes.
6. When the measurement is finished, click the "RECORDING DATA (CLICK TO STOP)" button followed by the "ACCEPT PLOT" button. Then turn off the laser for another 15 minutes to let it cool down for the next part of the experiment. You don't have to save the plot yet.

Complete these questions while waiting:

1. Calculate the frequency of the 633nm red light laser output.
2. Consider the laser tube to be a standing wave cavity of 25cm length. If we take the laser wavelength to be 0.5μm, what is the order m of the standing wave inside the cavity?
3. What is the frequency separation of adjacent longitudinal modes in the cavity? (standing waves with orders m and m + 1, where m is the order you calculated just now in point 2.)

4. The wavelength of light emitted by a moving atom will be Doppler shifted by an amount depending on its velocity relative to the observer. The distribution of velocities due to the thermal motion of the neon atoms in the laser's discharge tube gives rise to a broadening of 12 the spectral emission line such that its full width at half-maximum (FWHM) is given by the following equation:

$$\Delta_{V_D} = \frac{2}{\lambda}\sqrt{\frac{2kT \cdot \ln(2)}{M}} \tag{1}$$

$$\begin{aligned} \Delta_{V_D} &= \text{FWHM in units of frequency} \\ \lambda &= \text{central wavelength of spectral line} \\ M &= \text{mass of radiating atom} \\ T &= \text{temperature of gas} \end{aligned}$$

Assuming T 373K calculate the Doppler broadened linewidth of 632.8nm neon line. How many longitudinal modes would fit inside the Doppler boardened line? (Assume total width = 2· FWHM) Does the cavity oscillate in all of these modes?

Questions

1. What happens to the laser cavity with time? From this, explain the phenomena of mode sweeping and relate this back to your observed curve.

Polarised warm-up curve

1. Mount the linear polariser on the optical rail between the laser and the beam profiler, with the polariser set to 0 degrees. Align the laser beam through the polariser. Note which direction the numbers on the polariser are facing. Keep the polariser facing this way for Experiment 7 Longitudinal Modes.
2. In LabVIEW, toggle the "USING POLARISER" button to YES. Set your data acquisition time for 9 minutes.
3. Turn the laser on and immediately click the "CLICK TO START RECORDING DATA" button, rotating the polariser 10 degrees every 10 seconds for 9 minutes. Try and do this with your lab partner where one person keeps track of time and the other rotates the polariser.
4. When the measurement is finished, click the "RECORDING DATA (CLICK TO STOP)" button followed by the "ACCEPT PLOT" button. Both the unpolarised and polarised curves should now appear on the same plot. Save the graph and insert it into your lab book.

Questions

1. How does the polarisation of the laser beam vary with time?
2. From the experimental evidence you have, what do you think causes the periodic power oscillations? (Continue with the experiments for now and answer this question after completing Experiment 7 Longitudinal Modes)

   Note: Leave the laser on to stabilise for the rest of the experiments.

4.2.2 Experiment 7 - Longitudinal Modes

Reference 1 a) pg.591-594

Aim: *To measure the frequency separation of adjacent longitudinal modes of oscillation of the laser resonant cavity.*

A photodiode connected to a spectrum analyser (at end of bench) will be used to analyse the frequency separation between longitudinal modes. Note that the photodiode is a non-linear detector and hence a signal at the frequency difference between two longitudinal modes is detected. The spectrum analyser is NOT showing the frequency at which the longitudinal modes are oscillating in the laser cavity.

Method:

1. Mount the photodiode on the optical bench about 30cm from the laser output. Align the photodiode's sensitive area to the centre of the beam. (Note: The photodiode's sensitive area is the tiny 1mm black square in the centre of the photodiode.)
2. Place the polariser between the laser and the photodiode. Make sure the polariser is facing the same way as it did in section: *Polarised warm-up curve.*
3. Turn on the spectrum analyser and press the green "PRESET" button found to the left of the screen. (see the section below for more details on the spectrum analyser).
4. Rotate the angle of the polariser from 0 to 360 degrees and observe the change in height of the peak frequency. Note down the angles at which the peak frequency is at its maxima and minima. (if no peaks are seen during rotation, the photodiode is probably misaligned)

Spectrum Analyser

The correct settings should be activated when the green "Preset" button is pressed. If you find that you cannot observe any peak frequencies when rotating the polariser, set the spectrum analyser with the following settings:

1. The photodiode should be connected in parallel with a 50Ω load resistor to the spectrum analyser input. This is done using a T-connector at the spectrum analyser input.
2. Set the frequency span to 10MHz (using the "SPAN" button).
3. Set both resolution bandwidth (RBW) and video bandwidth (VBW) to 10kHz (both can be accessed using the "BW/Det" button).
4. Set the sweep time to 50ms (accessible from the "Sweep/Trig" button)
5. Under the amplitude menu ("AMPT" button), set the Ref Level to $50\mu V$, input attenuation to 0dB, scale type to linear, and turn on the RF preamp. You may change the Ref level if the height of the peak does not fit the screen.

Questions

1. From the frequency separation, calculate the effective length of the laser cavity.
2. Is the peak you observe on the spectrum analyser the only one that exists when you rotate the polariser? Explain your answer. If you think that other peaks exist, where would you find them?
3. Compare your results with section: Polarized warm-up curve and explain your observations. Note that the orientation of the modes from the laser may not be such that 0 degrees corresponds to the horizontal.

4.2.3 Experiment 8 - Beam Coherence

Reference 1 b) pg.560-573

Spatial Coherence

Aim: *To check whether the laser beam is coherent in phase across its cross-section.*

Method:

1. Remove all objects in the path of the laser beam and place the double slits in the slide holder using the translational stage, placing it at about 2m from the laser.
2. Slowly transverse the slits across the laser beam, noting the effect this has on the interference pattern observed on the screen at the end of the bench.

Questions

1. What property of the observed interference pattern tells you that the laser beam is spatially coherent?
2. Can you suggest another way to test the spatial coherence across the beam?

Temporal Coherence

Aim: *To determine the coherence length of the laser light by measuring the visibility of the interference fringes as a function of path difference in a Michelson interferometer.*

Note: A demonstrator's assistance may be required in setting up the interferometer.

Before commencing this experiment you should be familiar with the Michelson interferometer setup. (See Hecht pg.407-408 for more information)

Method:

1. Mount and lock the cross rail containing a fixed mirror, beam splitter and lens onto the rail, about 30 cm from the laser.
2. Adjust the beam splitter and the fixed mirror such that the laser reflected from the mirror passes through the lens and forms an expanded spot on the wall behind the lens.
3. Mount the moving mirror on the fixed rail at a distance from the beam splitter equal to the distance between the fixed mirror and the beam splitter.
4. Using the controls on the beam splitter and mirrors, adjust their tilt such that the beams from the fixed and moving mirrors overlap on the wall. Interference fringes should be visible on the wall at this point! If the fringes are moving too much, try to avoid touching or leaning on the lab bench.

You will now measure the visibility of the fringes as a function of the path difference between the fixed and moving mirrors.

1. Mount the Spiricon beam profiler at the end of the cross-rail and align the beams with the profiler.

2. From the desktop, open the LabVIEW program "PLL 8 - Temporal Coherence".

3. To start the program, click the "Run" button (white arrow in the top left). A window prompting for your name will appear. Enter your name, select where you wish to save your data then click "OK". The "BeamGauge" software should then initialise in the background.

4. Once the software has initialised (the "Ready" button should be visible) click the "CLICK TO START ACQUIRING DATA" button. If the detector is saturated, reduce the exposure by dragging the exposure marker to the left. The maximum pixel intensity should be less than 100. Adjust the position of the beams using the mirror and beam splitter controls so that the beam image is in the centre of the image box.

5. Observe the effect of vibrations on the fringes by giving the optical rail a quick light tap with your finger.

6. Place the moving mirror at a distance of 3cm from the the beam splitter. With the path difference equal to 0 when the fxed and moving mirrors are at the same distance from the beam splitter, a 3cm distance between beam splitter and moving mirror corresponds to a negative path difference. Enter this path difference in the "Path Diff" box to the right of the image box.

7. In LabVIEW, draw a line through the centre of the beam image.

8. Now block one of the beams and examine its intensity profile. If it looks Gaussian (compare to the blue Gaussian fit), click "ACCEPT GAUSS 1". Then do the same for the other beam and click "ACCEPT GAUSS 2".

9. Unblock both beams then click the "FIT INTERFERENCE" button. The value of the visibility of the fringes for the path difference will then be displayed. Click "ADD DATA POINT TO PLOT" then click "CLICK TO START ACQUIRING DATA" to re-initialise the detector.

10. Repeat steps 6-9 by moving the moving mirror in 3cm intervals along the optical rail until you reach a positive path difference of 30cm. Every time you move the mirror you should re-adjust the tilt slightly to overlap the two beams. Once finished, a plot of fringe visibility vs. path difference will be created.

11. Save your data by clicking the yellow "SAVE DATA" button. Insert both the plot and data into your lab book.

Questions

1. Explain the trend in the visibility vs. path difference plot. From the plot find the coherence length (with error) of the laser light. (Note: Coherence length = distance between successive visibility maxima (or minima).)

2. Compare the measured value of coherence length with the effective length of the laser cavity calculated in section Experiment 7: Longitudinal Modes. Comment on your results.

3. What would be the coherence length of a He-Ne laser operating in a single longitudinal mode if the typical width of one mode is ≈ 1 MHz?.